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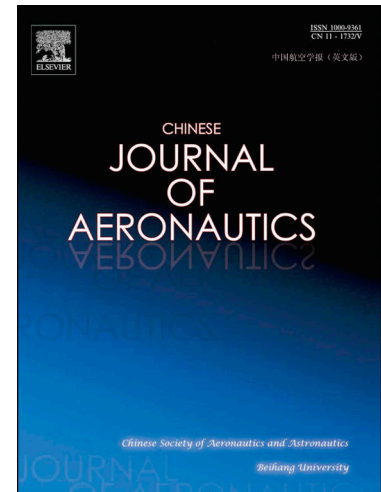
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# Modeling and simulation of aerial refueling multi-body system based on ALE-ANCF method

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## Abstract

The hose-drogue system is a common method for soft aerial refueling, whereby the refueling tanker tows the drogue through the hose. In this paper, a mathematical-physical model of the hose-drogue system is developed and simulated using the Absolute Nodal Coordinate Formulation (ANCF) finite element method. A numerical solution program based on ANCF and ALE-ANCF method was developed to simulate and analyze the horizontal and elongation release processes of the hose-drogue system at different towing points (underneath the wing and the belly of the aircraft). This program was developed by introducing an Arbitrary Eulerian-Lagrange (ALE) description. The numerical solution program, developed based on the ANCF and ALE-ANCF methods, represents a significant advancement in computational efficiency for the rigid-flexible coupled multibody system of the air refueling hose-drogue system. This program can provide a valuable reference for the qualitative design of the hose-drogue multibody system in soft air refueling, while maintaining the necessary accuracy.

**Keywords:** Arbitrary Eulerian-Lagrange (ALE); Absolute Nodal Coordinate Formulation (ANCF); aerial refueling; multibody system; rigid-flexible coupling

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## 1. Introduction

Air-to-air refueling technology, as a long-established and widely used method of aircraft fuel replenishment, plays an extremely important role in improving the sustained flight performance of aircraft. The air refueling of the recipient aircraft to obtain additional fuel can achieve a longer retention time and a larger mission activity radius (increased by 35% - 40%)<sup>1</sup>, multiple air refueling can greatly extend the retention time of the recipient aircraft, the recipient aircraft can reduce the amount of take-off fuel it carries in exchange for a heavier mission load or a shorter take-off distance, and by eliminating the process of landing to refuel and taking off again, aerial refueling technology reduces the total amount of fuel consumed by the aircraft per unit of time and improves mission economics. With the rapid development of aviation technology, the focus of future aerial refueling technology will gradually shift from manned refueling/receiving aircraft to manned-unmanned aircraft, unmanned-unmanned aircraft, and helicopter/rotorcraft aerial refueling. In these emerging application scenarios, the hose-drogue aerial refueling system still plays an irreplaceable role compared to the hard tube refueling system due to its smaller mass and size, suitable refueling flow rate, and high installation flexibility. As shown in Fig. 1, the refueling aircraft tows the drogue through the hose.

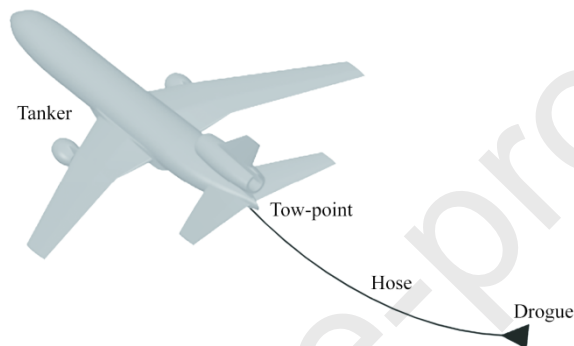


Fig. 1. Schematic diagram of soft air refueling system.

In recent years, there has been a growing need to predict and analyze the stable configuration and dynamic response characteristics of the rigid-flexible coupled structure of the hose-drogue system in the flight environment. This is particularly relevant in the design, research and development, and flight test and evaluation of the hose-drogue aerial refueling system. In comparison to flight tests and wind tunnel scaling tests, numerical simulation represents an efficient and cost-effective approach to research, offering a faster response time and the avoidance of scaling errors. To address the issue of dynamic modeling and simulation of hose-drogue devices for aerial refueling, a number of papers have proposed various methods for modeling the dynamic response of hose-drogue in the wake field of a refueling aircraft.

In order to investigate the dynamic characteristics of hose-drogue systems, many scholars and research institutes have conducted a large number of tests and modeling analyses. NASA Dryden Flight Research Center<sup>2,3</sup> has obtained abundant aerodynamic data, especially for modeling the hose-drogue dynamics, through wind tunnel tests and flight tests. Boeing<sup>4,5</sup> investigated the dynamic characteristics of the KC-10 hose whipping phenomenon through numerical simulations. Zhu and Meguid<sup>6,7</sup> proposed a new finite element formulation of a bending beam based on mechanics of materials to characterize the dynamics of the hose-drogue assembly. Ro et al.<sup>8</sup> tested the aerodynamic characteristics of the drogue through a wind tunnel and gave a kinetic model of the chain-link connection of the hose-drogue assembly. Zhao et al.<sup>9</sup> used the finite element analysis software ANSYS simulation and Simulink numerical computation to simulate the sinkage of drogue under smooth towing condition of hose. Cui et al.<sup>10</sup> investigated the main causes of Hose Whip Phenomenon (HWP) in hose-drogue and proposed effective preventive measures. Liu<sup>11</sup> established a coupling simulation framework and method for the tanker hose-drogue device based on Multi-Body Dynamics (MBD) and CFD by organically combining fluid-structure coupling analysis tools at multiple solution levels. Salehi Paniagua et al.<sup>12</sup> proposed a mathematical model containing different parameters for accurate modeling of hoses and drogues for aerial refueling. The above models have the following problems:

(1) Although they can reflect the dynamic characteristics of the hose-drogue assembly, the length of the hose is unchanged, and it is impossible to evaluate the dynamic characteristics of the system during the elongation and release

process.

(2) The hose-drogue structure is subjected to significant deformation and rotation under non-constant aerodynamic loads when it is towed by the fuel dispenser in a dynamic manner. When the traditional finite element method is employed to simulate the structural dynamics of the hose-drogue structure, a substantial number of finite element cells must be divided for discretization.

In this paper, the fore problems and limitations on modeling airborne towed hoses are broken by using Absolute Nodal Coordinate Formulation (ANCF)<sup>13-16</sup>. The method is capable of accurately modeling the large deformation problem of cables and beams in a small number of discrete cells, and is able to directly and uniformly solve the kinematic constraints of rigid-flexible coupled multibody dynamical systems. In addition, in order to solve the elongation-release problem of the hose-drogue system, the dynamics of this system is modeled using an ALE (Arbitrary Eulerian-Lagrange)-ANCF based method to solve the boundary-variable mass and length problems associated with the elongation-release process of the hose system. Finally, the efficiency of the model in a refueling scenario (hose-drogue system at different traction points for both horizontal and elongation release processes) is verified by numerical simulations.

## 2. Kinetic modeling of airborne towed system

This section focuses on modeling the dynamics of the hose-drogue system. Firstly, coordinate definition and model assumptions are carried out for the refueling system, based on which the hose unit is modeled by the ANCF method, while the boundary variable mass and length problems are described and completed using the ALE method, followed by the description of the multibody system and equation assembly. Aiming at the problem of solution accuracy and stability in the boundary variable mass and length, a mesh reconstruction method is given in meshing, and a linear interpolation method is given in node state processing. Finally, the wake flow field of the fuel dispenser is modeled, and the aerodynamic prediction methods for hose and drogue are given respectively.

### 2.1. Coordinate system definition and model assumptions

During the soft aerial refueling process, the refueling tanker always maintains a uniform level flight state, as shown in Fig. 2. Based on this, the Cartesian coordinate system  $Oxyz$  fixed on the refueling dispenser is identified as the inertial system, in which the point  $O$  is the drag release point of the fuel hose on the refueling dispenser,  $Ox$  is along the opposite direction of the horizontal flight of the refueling dispenser, the  $Oy$  axis points to the opposite direction of gravity, and the  $Oz$  axis is determined by the right-hand rule. In the aerial refueling multibody system, the fuel delivery hose  $OO_1$  is connected to the refueling tanker and the drogue with ball hinge at  $O$  and  $P$  respectively. The drogue is regarded as a rigid body with the center of mass at  $O_1$  and its body coordinate system at  $O_1x_1y_1z_1$ . In the initial state of release, the coordinate axes of  $O_1x_1y_1z_1$  and  $Oxyz$  are parallel to each other. Since bending and tension are the main deformation modes of the hose during the release-drag process, the refueling hose is modeled using an absolute nodal coordinate method capable of describing the bending and tension deformations<sup>17</sup>, and the Euler-Bernoulli beam assumption is used for this type of unit.

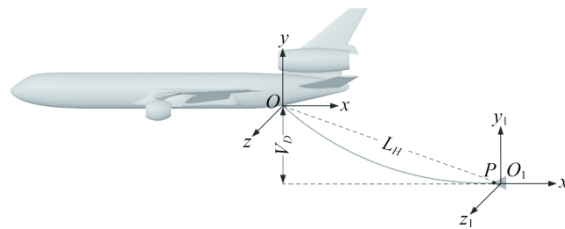


Fig. 2. Schematic of coordinate system definition and DDVP.

Also for analytical simplicity of the validation data for several elements of the hose and drogue system model, the vertical drogue position is expressed as a dimensionless parameter, called the Dimensionless Drogue Vertical Position (DDVP), which is used to represent the vertical position of the hose and drogue coupling with respect to the hose exit point of the system (Tow-point). The DDVP is defined as

$$\text{DDVP} = \frac{V_D}{L_H} \quad (1)$$

where  $V_D$  and  $L_H$ , as shown in Fig. 2, are the altitude difference between the center of mass of the drogue and Tow-point and straight-line distance from the Tow-point to the center of mass of the drogue, respectively.

## 2.2. Hose unit modeling

ANCF has the characteristics of constant mass matrix, good numerical stability and easy to combine with other multi-body systems, which has become one of the most effective modeling methods for multi-flexible body systems<sup>18-20</sup>, so the adoption of this method is extremely appropriate for modeling the dynamics of hoses. For the conventional absolute nodal coordinate formulation, it consists of two nodal tension cable units as shown in Fig. 3. Each node has six degrees of freedom, consisting of the  $3 \times 1$  position vector  $\mathbf{r}$  and the  $3 \times 1$  slope vector  $\mathbf{r}'$  tangent to the cable centerline. All degrees of freedom are defined in an absolute coordinate system. Fig. 3 shows a deformed flexural unit with arc length  $s$  at the top and an equivalent undeformed element with unstretched length  $L$  at the bottom. The nodes located at the ends of the elements are indicated by hollow circles.

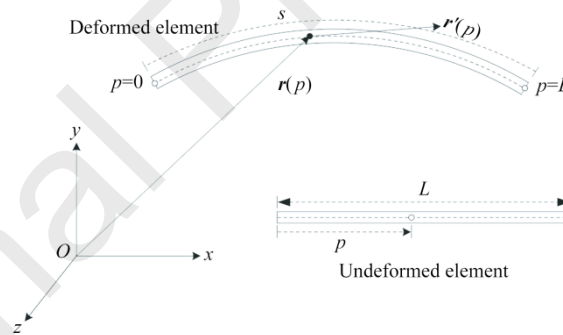


Fig. 3. Schematic diagram of ANCF unit.

For the elongation release problem of the air refueling hose-drogue system, which involves the change of the length and mass of the boundary unit, a typical schematic diagram of a one-dimensional moving medium unit is given in Fig. 4, in which  $p$  denotes the material coordinate of a material microcluster inside the unit, and the size of which is equal to the arc distance from the starting point of the medium to the material microcluster under the undeformed state of a one-dimensional medium,  $\mathbf{r}$  denotes the spatial coordinate vector of a material microcluster inside the unit,  $\mathbf{r}'$  The spatial slope vector of a material microcluster inside the unit,  $\mathbf{f}(p, t)$  is the external load force on the unit, the subscripts 1 and 2 indicate the physical quantities at the nodes of the unit respectively.

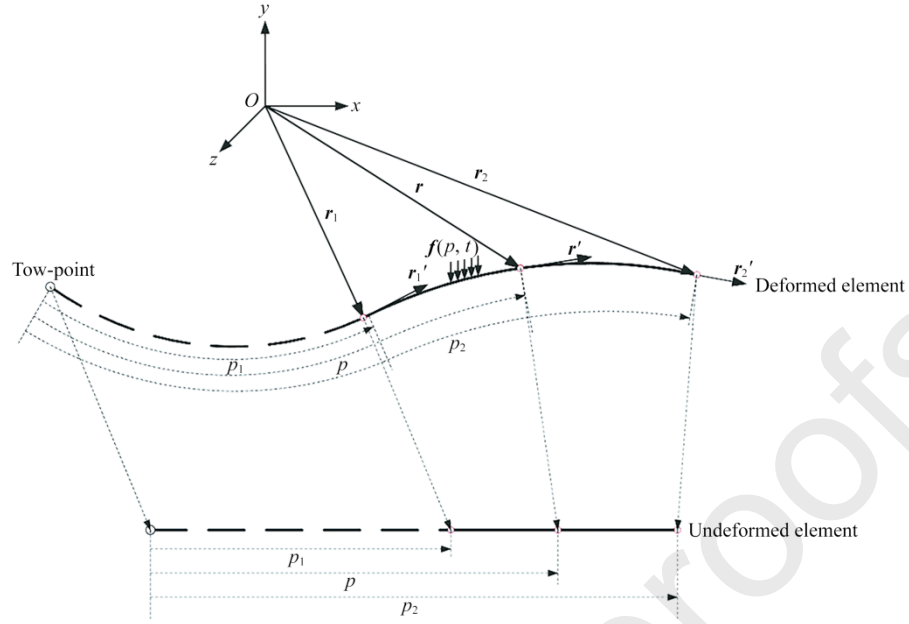


Fig.4. Schematic diagram of a typical one-dimensional moving medium unit.

Firstly, the generalized coordinates of the unit are defined, and according to the absolute node coordinate method, the spatial position, slope vector and material coordinates of the unit nodes are taken as the generalized coordinates, i.e.:

$$\mathbf{q} = [\mathbf{r}_1^T \quad (\mathbf{r}_1')^T \quad p_1 \quad \mathbf{r}_2^T \quad (\mathbf{r}_2')^T \quad p_2]^T \quad (2)$$

where  $\mathbf{r}$  is the spatial position of the unit node, and  $\mathbf{r}'$  is the derivative of the spatial coordinates of any material microcluster within the unit with respect to the unit material coordinates, i.e., the spatial slope of the unit axis.

$$\begin{aligned} \mathbf{r} &= [x \quad y \quad z]^T \\ \mathbf{r}' &= \frac{d\mathbf{r}}{dp} \end{aligned} \quad (3)$$

The vectors that define the composition of the grid coordinates for the traditional nodal coordinate method are as follows:

$$\mathbf{q}_e = [\mathbf{r}_1^T \quad (\mathbf{r}_1')^T \quad \mathbf{r}_2^T \quad (\mathbf{r}_2')^T]^T \quad (4)$$

In order to describe the motion of any point inside the cell, the Hermite interpolation function is introduced with:

$$\mathbf{r}(p) = s_1 \mathbf{r}_1 + s_2 \mathbf{r}_1' + s_3 \mathbf{r}_2 + s_4 \mathbf{r}_2' \quad (5)$$

The expressions  $s_1 - s_4$  in Eq.(5) are respectively:

$$\begin{cases} s_1 = \frac{(p-p_2)^2(2p-3p_1+p_2)}{(p_2-p_1)^3}, & s_2 = \frac{(p-p_1)(p-p_2)^2}{(p_1-p_2)^2} \\ s_3 = \frac{(p-p_1)^2(2p+p_1-3p_2)}{(p_1-p_2)^3}, & s_4 = \frac{(p-p_2)(p-p_1)^2}{(p_1-p_2)^2} \end{cases} \quad (6)$$

The following intermediate variables are introduced to facilitate the derivation:

$$\mathbf{S}_e = [s_1 \mathbf{I} \quad s_2 \mathbf{I} \quad s_3 \mathbf{I} \quad s_4 \mathbf{I}] \quad (7)$$

where  $\mathbf{I}$  is the  $3 \times 3$  unit matrix and  $\mathbf{S}_e$  is a cubic function. This yields that any point inside the unit has:

$$\mathbf{r}(p) = \mathbf{S}_e \mathbf{q}_e \quad (8)$$

The cell real-time length is given by the difference between the material coordinates at the two nodes:

$$l_e = p_2 - p_1 \quad (9)$$

It should be noted that  $\mathbf{r}$ ,  $\mathbf{r}'$  and  $p$  in the above equations are the observed quantities obtained at the nodes of the cell, not the physical quantities tracking the material points, and the shape function of the cell constrains the transverse motion of the medium, but does not constrain the motion of the medium along the axis of the cell. The rates of change of the matter coordinates at the cell nodes are  $\dot{\mathbf{p}}_1$ ,  $\dot{\mathbf{p}}_2$ . If  $\dot{\mathbf{p}}_2 - \dot{\mathbf{p}}_1 > 0$  then the cell is in elongation and vice versa. If  $\dot{\mathbf{p}}_2 = \dot{\mathbf{p}}_1 \neq 0$ , then the length of the cell does not change, but there is matter transport inside the cell.

Additionally, the velocity of a point on the cable is

$$\dot{\mathbf{r}} = \mathbf{S}_e \dot{\mathbf{q}}_e + \left( \frac{\partial \mathbf{S}_e}{\partial p_1} \dot{\mathbf{p}}_1 + \frac{\partial \mathbf{S}_e}{\partial p_2} \dot{\mathbf{p}}_2 \right) \mathbf{q}_e = \mathbf{S} \mathbf{q} \quad (10)$$

where  $\mathbf{S}$  is the form function with the expression:

$$\mathbf{S} = \begin{bmatrix} s_1 \mathbf{I} & s_2 \mathbf{I} & \frac{\partial \mathbf{S}_e}{\partial p_1} \mathbf{q}_e & s_3 \mathbf{I} & s_4 \mathbf{I} & \frac{\partial \mathbf{S}_e}{\partial p_2} \mathbf{q}_e \end{bmatrix} \quad (11)$$

It is clear that if there is no material transport at the cell nodes, i.e.  $l_e = \text{constant}$  or  $\dot{\mathbf{p}}_2 = \dot{\mathbf{p}}_1$  and  $\dot{\mathbf{p}}_2 = \dot{\mathbf{p}}_1$ , then Eq. (10) degenerates to a velocity expression for a conventional dielectric cell.

According to the control equation proposed by Hong <sup>16</sup>:

$$\begin{cases} \mathbf{M}\ddot{\mathbf{q}} + \mathbf{Q}_p + \mathbf{Q}_e + \mathbf{Q}_f + \mathbf{C}_q^T \boldsymbol{\lambda} = \mathbf{0} \\ \mathbf{C}(\mathbf{q}, t) = \mathbf{0} \end{cases} \quad (12)$$

where  $\mathbf{M}$  is the cell mass matrix,  $\mathbf{Q}_p$  is the generalized force representing the cable inertia force due to changes in the material coordinates  $p_1$  and  $p_2$  i.e. additional inertia force,  $\mathbf{Q}_e$  represents the generalized force of the internal elastic potential,  $\mathbf{Q}_f$  represents the externally applied force, and  $\mathbf{C}$  represents the complete set of constraints. The forces required to satisfy the constraints are given by the product of the constraint Jacobian matrix  $\mathbf{C}_q^T$  and the Lagrange multiplier  $\boldsymbol{\lambda}$  vector.

The unit quality matrix  $\mathbf{M}_e$  is given by the following equation:

$$\mathbf{M}_e = \int_{p_1}^{p_2} \rho A \mathbf{S}^T \mathbf{S} dp \quad (13)$$

where  $\rho$  is the hose density and  $A$  is the cross-sectional area. The additional inertia force  $\mathbf{Q}_p$  is

$$\mathbf{Q}_p = \int_{p_1}^{p_2} \rho A \mathbf{S}^T \ddot{\mathbf{p}} dp \quad (14)$$

Generalized elastic force  $\mathbf{Q}_e$  for:

$$\mathbf{Q}_e = \int_{p_1}^{p_2} EA \left( \frac{\partial \varepsilon}{\partial \mathbf{q}} \right)^T \varepsilon dp + \frac{1}{2} \int_{p_1}^{p_2} EJ \left( \frac{\partial \kappa}{\partial \mathbf{q}} \right)^T \kappa dp \quad (15)$$

where  $\varepsilon$  is the longitudinal strain,  $\kappa$  is the elemental curvature, and  $E$  is the elastic modulus of the material.  $J$  is the cross-sectional moment of inertia. Since the cross-section of the one-dimensional medium has been previously assumed to be circular, the cross-sectional moment of inertia has rotational symmetry and there is no need to take into account the change in the pointing of the main axis of inertia at the interface. In the present study, the strain and curvature are quoted from Westin and Irani<sup>21</sup>, defined as follows:

$$\begin{aligned} \varepsilon &= (\mathbf{r}_p^T \mathbf{r}_p - 1) \\ \kappa &= |\mathbf{r}_{pp}| \end{aligned} \quad (16)$$

For an external distributed force  $\mathbf{f}(p)$  having a corresponding generalized force, the generalized force vector  $\mathbf{Q}_f$  can be determined by pre-multiplying the transpose of the shape function and integrating over the length of the element if the distributed force  $\mathbf{f}(p)$  is given per unit length<sup>22</sup>:



$$\mathcal{Q}_f = -\int_{p_1}^{p_2} \mathcal{S}^T f(p) dp \quad (17)$$

### 2.3. Multibody system description and multibody dynamical equation assembly

With the development of computers and numerical methods, multi-body system dynamics has experienced rapid development in the past decades. Rigid-body dynamics has now reached a relatively mature stage. Adopting quaternions as the generalized coordinates describing the rotation of a rigid body, the Lagrange equations can be used to establish the dynamical equations of rigid bodies and multi-body systems with complete constraints<sup>23</sup>:

$$\begin{cases} m_b \ddot{\mathbf{r}}_b - \mathbf{F}_b = \mathbf{0} \\ 4\mathbf{G}_b^T \mathbf{J}_b \mathbf{G}_b \dot{\boldsymbol{\theta}}_b - 8\mathbf{G}_b^T \mathbf{J}_b \mathbf{G}_b \boldsymbol{\theta}_b + 2\boldsymbol{\theta}_b \sigma_b - \mathcal{Q}_b = \mathbf{0} \\ \boldsymbol{\theta}_b^T \boldsymbol{\theta}_b - 1 = 0 \end{cases} \quad (18)$$

where  $\mathbf{r}_b$  is the coordinates of the center of mass of the rigid body,  $\boldsymbol{\theta}_b$  is the Euler quaternion of the rigid body,  $m_b$  and  $\mathbf{J}_b$  are the mass and inertia matrices of the rigid body, respectively,  $\mathbf{F}_b$  and  $\mathcal{Q}_b$  are the generalized external force and the generalized external moment of the rigid body, and  $\mathbf{G}_b$  is the matrix composed of quaternions, denoted as

$$\mathbf{G}_b = \begin{bmatrix} -\theta_{b,1} & \theta_{b,0} & \theta_{b,3} & -\theta_{b,2} \\ -\theta_{b,2} & -\theta_{b,3} & \theta_{b,0} & \theta_{b,1} \\ -\theta_{b,3} & \theta_{b,2} & -\theta_{b,1} & \theta_{b,0} \end{bmatrix} \quad (19)$$

In addition, the connection between the hose and the drogue is characterized as a vertical constraint, which is shown schematically in Fig. 5, so the constraint equation for both is expressed as

$$\mathbf{r}_n'^T (\mathbf{R}\mathbf{h}) = 0 \quad (20)$$

where  $\mathbf{r}_n'$  is the spatial slope vector of the  $n$ th node corresponding to the constraint on the hose,  $\mathbf{R}$  is the coordinate transformation matrix of the drogue, and  $\mathbf{h}$  is the projection of the direction vector corresponding to the constraint on the drogue in the center-of-mass coordinate system.

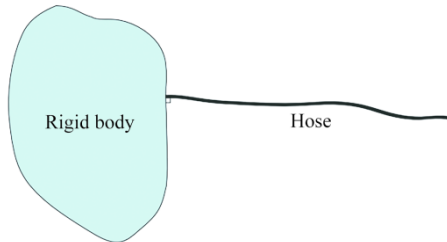


Fig. 5. Schematic diagram of rigid medium constraints<sup>16</sup>.

Combining Eq. (12), Eq. (18) and Eq. (20), the general form of the dynamical equations of the multibody system containing the release process of the hose drogue is assembled to obtain the general form of the dynamical equations of the multibody system containing the release process of the hose-drogue:

$$\begin{cases} M\ddot{\mathbf{q}} + \mathbf{Q} + \mathbf{C}_q^T \boldsymbol{\lambda} = \mathbf{0} \\ \mathbf{C}(\mathbf{q}, t) = \mathbf{0} \end{cases} \quad (21)$$

where  $\mathbf{M}$  and  $\mathbf{Q}$  denote the overall generalized mass array and generalized force vectors of the system,  $\mathbf{C}$  denotes the overall constraint equations of the system,  $\mathbf{C}_q$  denotes the constraint Jacobi matrices of the system.

#### 2.4 Flow field modeling and combined external force calculation

According to the assumptions in Section 2.1, the flow field behind the refueling tanker can be approximated as a straight homogeneous flow superimposed on the tail vortex field of the tanker, since the inertial coordinate system  $Oxyz$ , which is solidly attached to the refueling tanker, is used to measure all the physical quantities during the aerial refueling process. Existing tail vortex flow field models include Rankine vortex model, adapted vortex model, Lamb-Oseen vortex model, Smooth blending vortex model and Hallock-Burnham vortex model, etc.<sup>24-25</sup>. The Hallock-Burnham vortex model is usually used to calculate the velocity field generated by the tail vortex in the simulation calculation of air refueling<sup>26, 27</sup>.

##### 2.4.1 Calculation of combined external forces of hoses

For the air refueling system, the combined external force  $\mathbf{F}_e$  acting on the hose unit consists of the gravitational force  $\mathbf{G}_e$  and the aerodynamic force  $\mathbf{F}_{ea}$ , which is expressed as

$$\mathbf{F}_e = \mathbf{G}_e + \mathbf{F}_{ea} \quad (22)$$

where the unit gravity  $\mathbf{G}_e$  is:  $\mathbf{G}_e = m_e \mathbf{g}$ ,  $m_e$  is the hose unit mass, and  $\mathbf{g} = [0 \ 0 \ -9.81]^T \text{ m/s}^2$  is the gravity acceleration vector.

The hose aerodynamic force is based on the model used by Driscoll and Nahon<sup>28</sup> and Buckham et al.<sup>29</sup>, which is for underwater towing, considering the difference between airborne towing and underwater towing, and also for the aerial refueling problem, the hose-drogue system is also affected by the atmospheric turbulence and the refueling tanker wake field, it is worthwhile to point out that this paper only takes the refueling tanker wake field into consideration, therefore, in the reference of the model, we make some modifications in the components of the tow force as mentioned above. Therefore, the components of the towing force of the model are modified, and the final aerodynamic components are obtained as follows:

$$\begin{cases} \mathbf{F}_{D,n} = -\frac{1}{2} \rho_f d C_d |\mathbf{V}|^2 \frac{\mathbf{V}_n}{|\mathbf{V}_n|} \\ \mathbf{F}_{D,t} = -\frac{1}{2} \rho_f d C_f |\mathbf{V}|^2 \frac{\mathbf{V}_t}{|\mathbf{V}_t|} \text{sgn}(\mathbf{V}_t \cdot \mathbf{u}_t) \end{cases} \quad (23)$$

where  $d$  is the outer diameter of the hose,  $\rho_f$  is the density of air,  $\mathbf{V} = \mathbf{V}_f - \mathbf{u}$ ,  $\mathbf{V}_f$  are the local incoming velocity under the joint influence of the incoming velocity and the wake flow field of the refueling pump,  $\mathbf{u}$  is the hose velocity,  $V_n$  and  $V_t$  are the normal and tangential components of  $\mathbf{V}$ , respectively. In addition,  $C_d$  and  $C_f$  are the coefficients of differential pressure and friction respectively, and their values were obtained using the experimental formula<sup>30,31</sup>.  $V_n$  and  $V_t$  are given by the following equation:

$$\begin{cases} V_t = (\mathbf{V}_t \cdot \mathbf{u}_t) \mathbf{u}_t \\ V_n = \mathbf{V} - V_t \end{cases} \quad (24)$$

where  $\mathbf{u}_t = \mathbf{r}_p / |\mathbf{r}_p|$  is the unit tangential vector along the centerline of the hose. Fig. 6 illustrates a schematic of the absolute and relative flow velocity vectors and their components normal and tangential to the hose unit. The local incoming flow velocity vector  $\mathbf{V}_f$  is represented as a solid red line and the relative flow velocity vector  $\mathbf{V}$  is represented as a solid blue line. The normal  $V_{f,n}$  and tangential  $V_{f,t}$  components of the incoming velocity and the normal  $V_n$  and tangential  $V_t$  components of the relative velocity are shown as dashed lines.

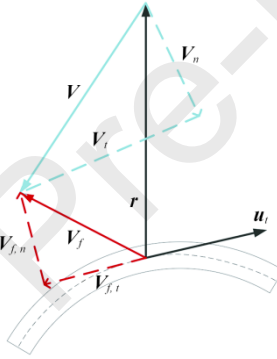


Fig. 6. Schematic representation of absolute and relative flow velocity vectors and their components in the normal and tangential directions of hose unit.

#### 2.4.2 Calculation of combined external forces of drogue

The combined external force  $\mathbf{F}_d$  acting on the center of mass of the drogue consists of the gravitational force  $\mathbf{G}_d$  and the aerodynamic force  $\mathbf{F}_{da}$ , which is expressed as

$$\mathbf{F}_d = \mathbf{G}_d + \mathbf{F}_{da} \quad (25)$$

where the gravitational force is:  $\mathbf{G}_d = m_d \mathbf{g}$  and  $m_d$  is the mass of the drogue. The expression for the aerodynamic force  $\mathbf{F}_{da}$  is

$$\mathbf{F}_{da} = \frac{1}{2} \rho_f A_d C_{\text{drogue}} |\mathbf{V}_f - \mathbf{V}_d| (\mathbf{V}_f - \mathbf{V}_d) \quad (26)$$

where  $A_d$  is the characteristic drag area of the drogue,  $C_{\text{drogue}}$  is the aerodynamic drag coefficient of the drogue, the size of which is taken as 0.8 with reference to the value in Ref. 32, and  $V_d$  is the center of mass velocity of the drogue.

### 3. Parameter and numerical solution program

#### 3.1 Parameter control of variable length hose unit

##### 3.1.1 Cell length control

During the release process of the refueling hose drag system, the length of the cell near the drag point changes with time due to the time-varying characteristics of the hose length. Obviously, if the cell length is too long, it is equivalent to the finite element mesh density decrease, resulting in the decrease of discretization accuracy, if the cell length is too short, the cell mass array and cell stiffness array obtained by cell integration are close to zero, and together with the influence of numerical truncation error, it may directly lead to the condition number of these matrices is too large, which makes the numerical computation fail. Therefore, in the actual simulation, the cell length must be controlled to be within a reasonable range.

The dynamic meshing method<sup>16</sup> is used to control the cell length in real time, and the execution process is shown in Fig. 7. During the computation process, the cell length around the sliding node is detected in real time, and if the cell length is out of the range of the threshold value, appropriate processing is required. Specifically, when the unit length is more than 1.5 times the reference length, a Lagrange node is inserted into the unit, and the original unit is divided into two new units, and the new unit away from the drag point is a constant length unit, and the unit length is equal to the reference length, and the unit near the tow-point (shown in Fig. 8) is a variable length unit, and the initial unit length is the original unit length minus the reference length.

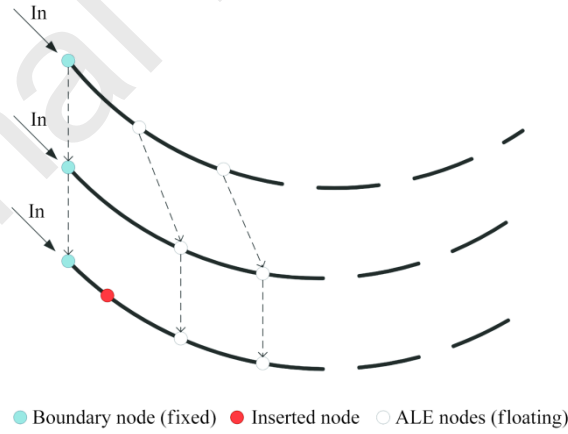


Fig.7. Schematic diagram of dynamic meshing process.

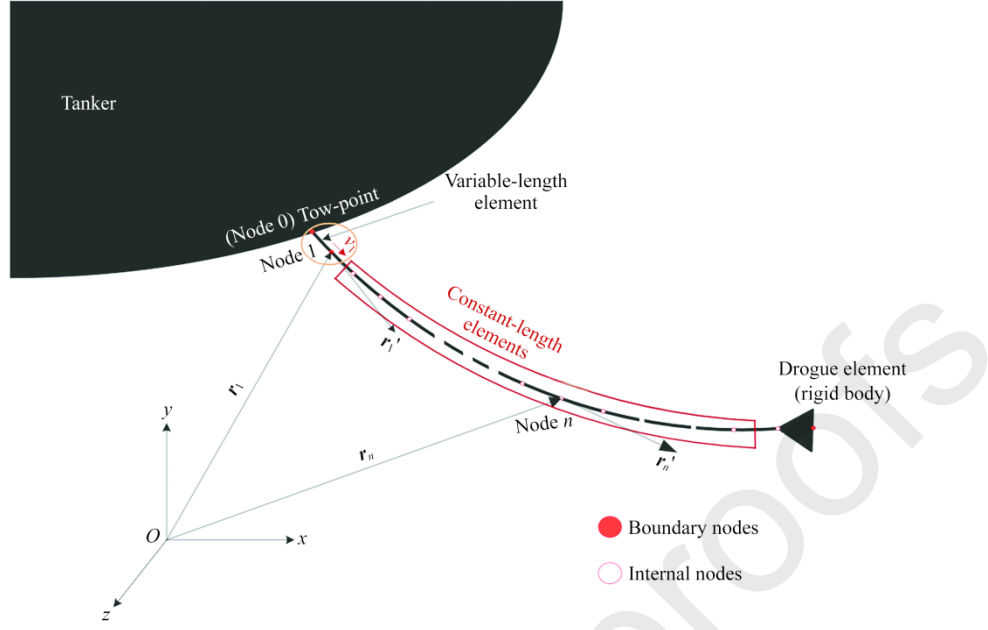


Fig. 8. Schematic diagram of release process of refueling hose-drogue system.

### 3.1.2 Nodal parameter interpolation

Due to the node insertion, it is necessary to interpolate each kinematic parameter of the newly added node, considering the numerical solution accuracy and stability, a linear interpolation method is proposed. From Eq. (9), the real-time length of the unit is the difference of the node material coordinates  $l_e$ , defining the reference length of the unit as  $l_0$ , and introducing the intermediate variable  $k$  for the convenience of derivation, whose physical meaning is the ratio of the length of the new unit from the drag point to the new node to the length of the unit when the node is not inserted, and the expression is

$$k = \frac{l_e - l_0}{l_e} \quad (27)$$

To ensure the stability and efficiency of the numerical solution, the state at the new node is obtained by linear interpolation. Let the coordinate of the drag point be  $r_a$  and the coordinate of another node located in the same cell as the drag point be  $r_b$ , and then the expression of the coordinate of the newly inserted node  $r_{b\_new}$  is obtained as follows:

$$r_{b\_new} = k(r_a + r_b) + r_a \quad (28)$$

### 3.2 Numerical solution program

All physical properties and geometric data for the given hose-drogue system and tanker are shown in Table 1. From Section 2.1, the towing point is the inertial coordinate system, and given the initial state of the hose-drogue, a generalized  $\alpha$ -method can be used to solve the multibody dynamics problem, including the ANCF and ALE-ANCF

models. The method has the same time-step equations as the Newmark method, and Eq. (29) shows the two time-step equations of the Newmark method. Fig. 9 shows the flowchart of the numerical integration solution using the generalized  $\alpha$ -method.  $h$  is the time step,  $\gamma$  and  $\beta$  are the integration parameters associated with the generalized  $\alpha$ -method respectively.

It is worth mentioning that during the elongation and release process of the hose-taper sleeve system, the mass of the boundary cells changes with time, so the mass matrix involved in the solution process is a very mass matrix. In order to solve this problem, the specific numerical solution method used in this study can be found in Ref. 33.

$$\begin{cases} \mathbf{\ddot{q}}_{n+1} = \mathbf{\ddot{q}}_n + (1-\gamma)h\mathbf{\ddot{q}}_n + \gamma h\mathbf{\ddot{q}}_{n+1} \\ \mathbf{q}_{n+1} = \mathbf{q}_n + h\mathbf{\dot{q}}_n + \frac{h^2}{2}[(1-2\beta)\mathbf{\ddot{q}}_n + 2\beta\mathbf{\ddot{q}}_{n+1}] \end{cases} \quad (29)$$

Table 1 Parameters related to hose-drogue systems and tanker.

| Component |                                 | Configuration |
|-----------|---------------------------------|---------------|
| Tanker    | Weight(kg)                      | 20 552        |
|           | Span, m                         | 11.43         |
|           | Reference area(m <sup>2</sup> ) | 37.16         |
| Hose      | Length(m)                       | 14.325 6      |
|           | Diameter (internal)(m)          | 0.067 3       |
|           | Diameter (external)(m)          | 0.050 8       |
|           | Weight/Length(kg/m)             | 4             |
|           | Elastic modulus(MPa)            | 265           |
| Drogue    | Weight(kg)                      | 29.5          |
|           | Reference area(m <sup>2</sup> ) | 0.282 6       |

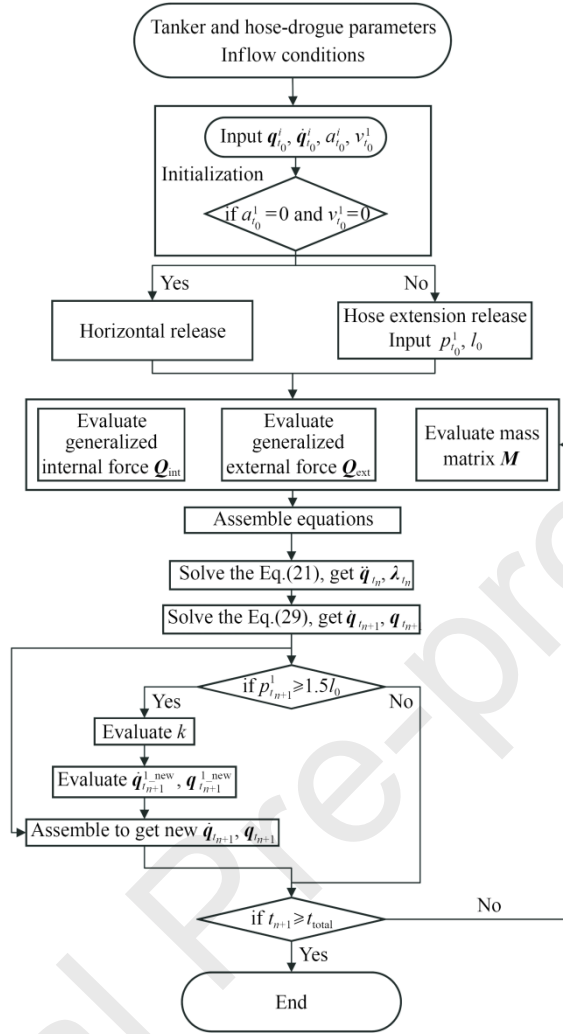


Fig. 9. Flowchart of numerical solution.

#### 4. Simulations and results

A MATLAB script simulation program was developed based on the mathematical-physical model presented in the previous section, and all the calculations were done on an Intel Core i7-8700 processor with 32 GB of memory. Numerical integration of the equations of motion with the multi-body dynamics equations using the generalized alpha method is performed, and the final solution is obtained to obtain the kinematic state of the hose-drogue system. In this section, the simulation results are shown and the published flight test experimental values and previous numerical simulation results are compared with the numerical simulation results in this paper.

##### 4.1 Equilibrium position analysis of hose drogue system

In the actual aerial refueling scenarios, two modes of under-wing refueling and belly refueling are usually involved, and the two refueling modes will affect the lateral motion of the system, so this section will simulate the two modes, and the initial state of the hose-drogue for both modes is assumed to be placed horizontally. Since the steady state position of the hose-drogue under different flight conditions is crucial for safe aerial refueling. A schematic diagram of the two refueling methods is shown in Fig. 10.



Fig.10. Schematic diagram of refueling method.

#### 4.1.1 Equilibrium position analysis using abdominal refueling

For all cases the solution time step is 0.01, the total solution time is 80 s, and the system reaches steady state in 40 s, with an average CPU computation time of 76 s per case. Table 2 shows the computed steady state values of the hose-drogue system under typical refueling conditions, in addition this paper compares the results with the published flight test results<sup>2-3</sup> and the simulation results from the Ref. 29 as shown in Fig. 11. In Fig. 11, 1 kt=1.852 km/h.

Table 2 Calculated steady state values of DDVP for hose-drogue system under typical refueling conditions.

| Steady-state values  | Altitude $H(\text{m})$ |         |         |         |
|----------------------|------------------------|---------|---------|---------|
|                      | 2 286                  | 3 048   | 7 620   | 9 144   |
| $V = 170 \text{ kt}$ | 0.460 3                | 0.482 8 | 0.608 3 | 0.634 5 |
| $V = 190 \text{ kt}$ | 0.401                  | 0.418 8 | 0.561 8 | 0.601 1 |
| $V = 210 \text{ kt}$ | 0.35                   | 0.362 5 | 0.509 5 | 0.556 6 |
| $V = 230 \text{ kt}$ | 0.305 6                | 0.319   | 0.457 4 | 0.509   |
| $V = 250 \text{ kt}$ | 0.270 4                | 0.282 6 | 0.408 8 | 0.461 2 |
| $V = 270 \text{ kt}$ | 0.241 1                | 0.252 2 | 0.365 7 | 0.416 2 |
| $V = 290 \text{ kt}$ | 0.213 7                | 0.228 3 | 0.327 7 | 0.375 7 |
| $V = 310 \text{ kt}$ | 0.190 6                | 0.203 8 | 0.294 9 | 0.339 9 |



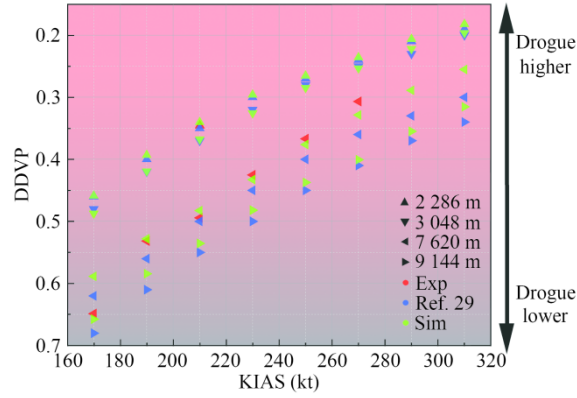


Fig. 11. Comparison of DDVP calculated values with Ref.29 and flight test values.

Fig. 11 shows a comparison of the DDVP values obtained from the calculations in this paper with the NASA flight test data and the data obtained from the Kamman et al.<sup>29</sup> simulation in the case of belly refueling. The simulated values at a flight speed of 210 kt and an altitude of 2 286 m are in general agreement with the flight test data and the simulation by Kamman et al.<sup>29</sup> As can be seen from Fig. 11, the flight test data are mainly concentrated at an altitude of 7 620 m. From this figure it can be clearly seen that the data calculated in this paper are closer to the flight test values than the data simulated by Kamman et al.<sup>29</sup> This indicates that the methodology used in this paper has a smaller error at high altitude conditions. In addition, at the same altitude, the DDVP values show a linear change with the change in flight speed, and the trend of the results of this paper and the test values are basically in agreement with the simulation values of Kamman et al.<sup>29</sup> This result demonstrates the credibility of the methodology used in this paper.

Fig. 12 and Fig. 13 show the equilibrium state of the system by belly refueling at different altitudes with the same flight speed (250 kt) and at the same altitude (2 286 m) with different flight speeds, it clearly shows the effect of speed and altitude on the system. Fig. 14 shows the equilibrium process diagram of the system with time at flight speed of 250 kt and altitude of 2 286 m, Figs. 15-17 show the variation of each parameter of the drogue obtained from the calculations with the advancement of time, the center of mass of the drogue is finally stabilized at (13.85, -3.8)m, and the magnitude of the pitch angle of the drogue is about 9°.

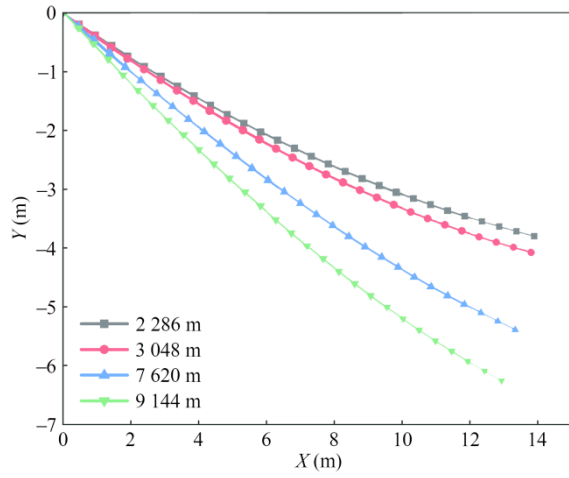


Fig. 12. Equilibrium state of system for different altitudes with  $V=250$  kt.

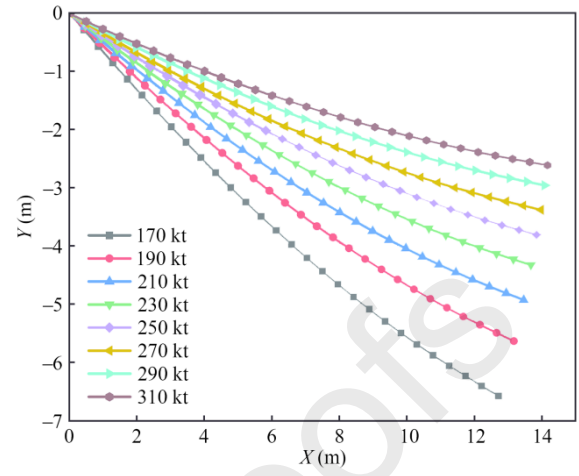


Fig. 13. Equilibrium state of system for different flight speeds with  $H=2\ 286$  m.

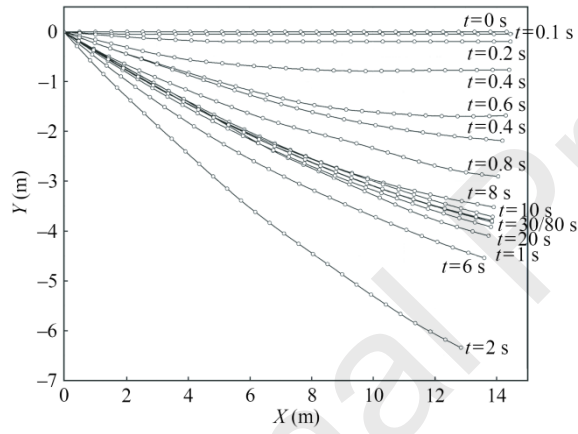


Fig. 14. Equilibrium process of system with  $V=250$  kt,  $H=2\ 286$  m.

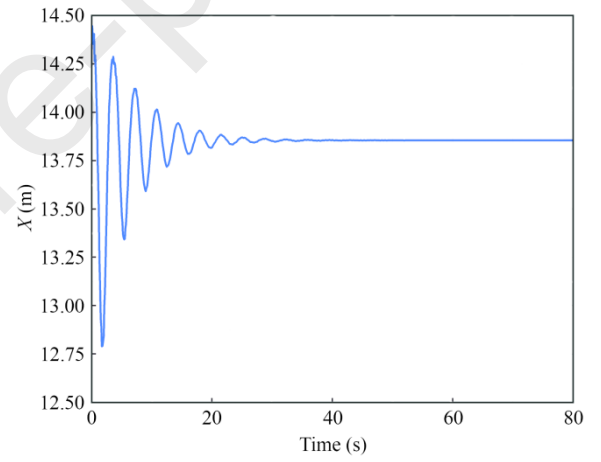


Fig. 15. Trajectory of the center of mass of drogue in  $X$ -direction ( $V=250$  kt,  $H=2\ 286$  m).

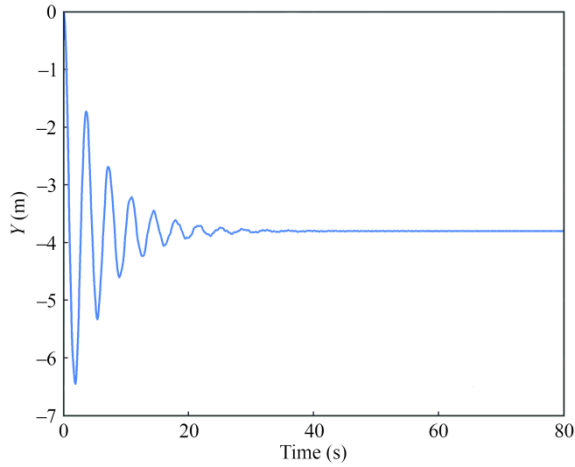


Fig. 16. Trajectory of the center of mass of drogue in Y-direction( $V=250$  kt,  $H=2\ 286$  m).

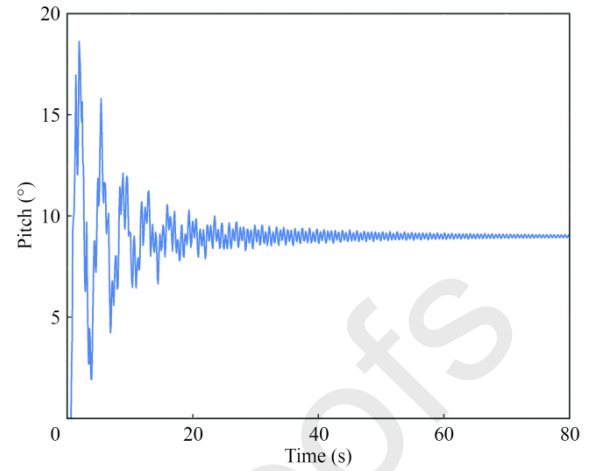


Fig. 17. Variation of pitch angle of drogue( $V=250$  kt,  $H=2\ 286$  m).

#### 4.1.2 Balance position analysis using under-wing refueling

The main content of this section is to analyze the balance position of the hose drag system when the underwing refueling method is used, and the hose drag point in this example is at the center wing.

Fig. 18 and Fig. 19 show the equilibrium state of the system with under-wing refueling at different altitudes with the same airspeed (170 kt) and at the same altitude (2 286 m) with different airspeeds, from which the lateral effects of airspeed and altitude on the system can be clearly observed. The time plot of the center of mass trajectory of the cone at flight speed 170 kt and altitude 2 286 m is given in Fig. 20, from which the trajectory plot of the cone can be clearly obtained, which has uncertainty. Fig. 21 shows the spatial position diagram of the system and the projection in three planes when the system is in stable equilibrium, and the center of mass of the drogue is finally stabilized at (12.75, -6.25, 1.75)m by combining Figs. 22-24. Figs. 25-27 show the changes in the attitude angle of the cone obtained from the calculation with time, and its pitch angle is about  $17^\circ$ , yaw angle is about  $16^\circ$ , and roll angle is about  $-7.8^\circ$  at the time of stabilization.

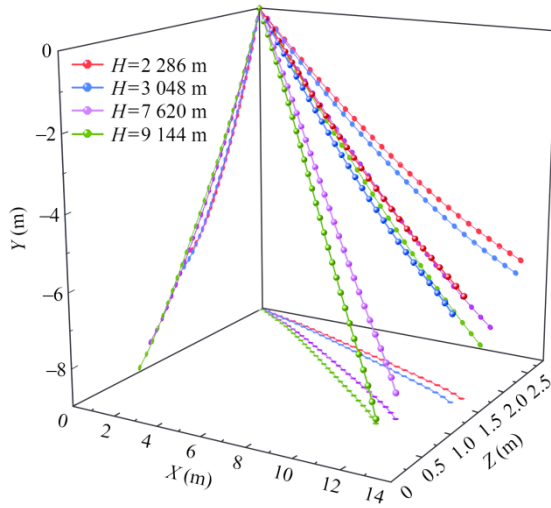


Fig. 18. Equilibrium state of system for  $V=170$  kt and different altitudes.

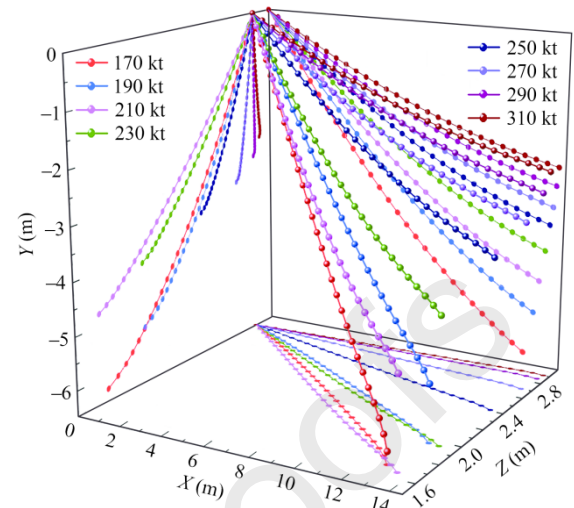


Fig. 19. Equilibrium state of system for  $H=2286$  m and different flight speeds.

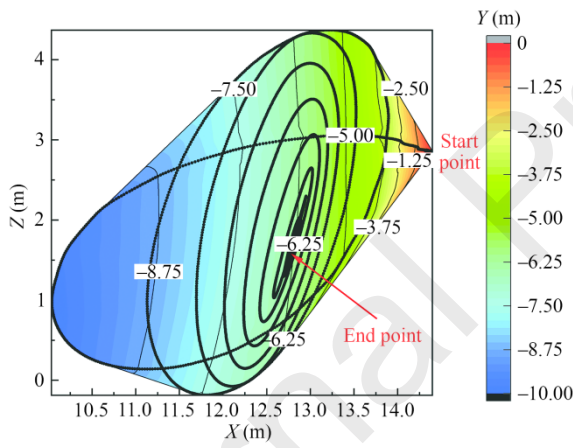


Fig. 20. Time course of the center of mass trajectory of conical for  $V=170$  kt,  $H=2286$  m.

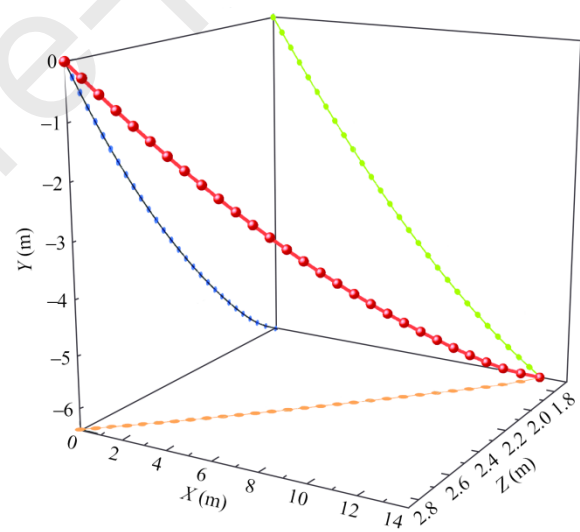


Fig. 21. Equilibrium state of system for  $V=170$  kt,  $H=2286$  m.

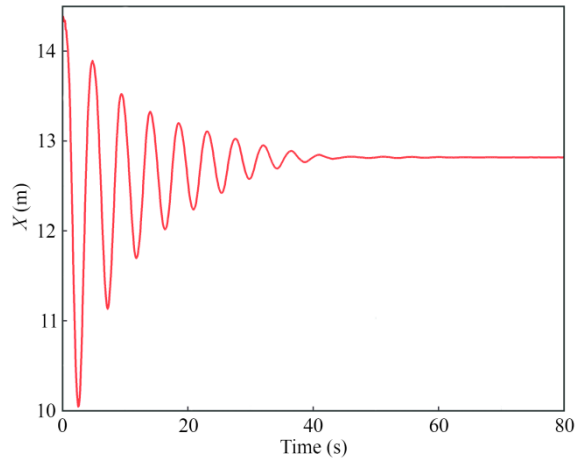


Fig. 22. Trajectory of the center of mass of the drogue in  $X$ -direction( $V=170$  kt,  $H=2\ 286$  m).

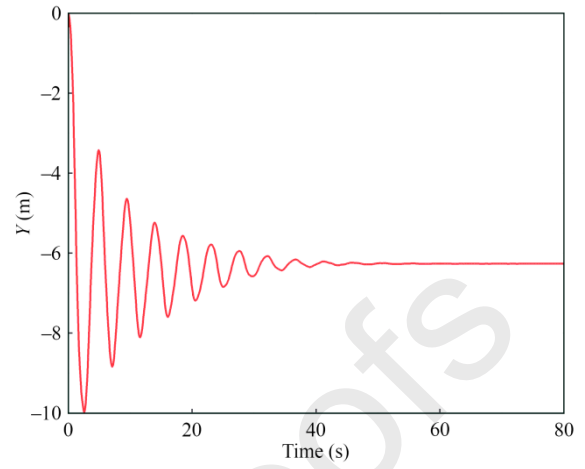


Fig. 23. Trajectory of the center of mass of the drogue in  $Y$ -direction( $V=170$  kt,  $H=2\ 286$  m).

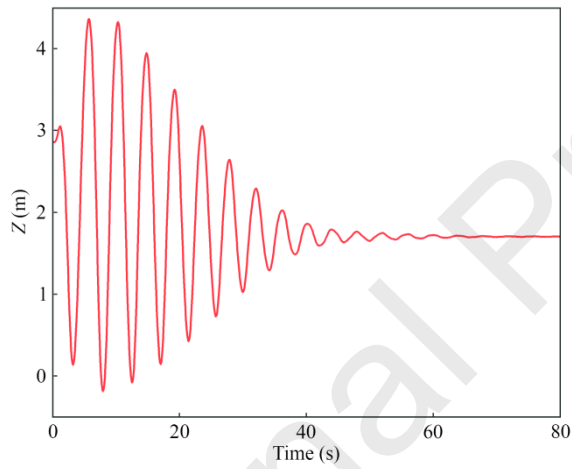


Fig. 24. Trajectory of the center of mass of drogue in  $Z$ -direction( $V=170$  kt,  $H=2\ 286$  m).

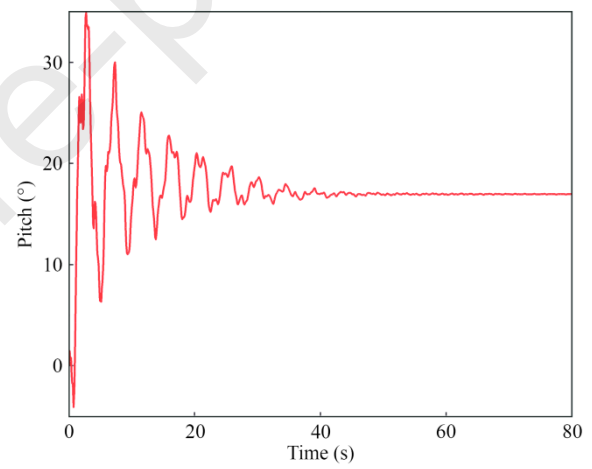


Fig. 25. Variation of the pitch angle of drogue( $V=170$  kt,  $H=2\ 286$  m).

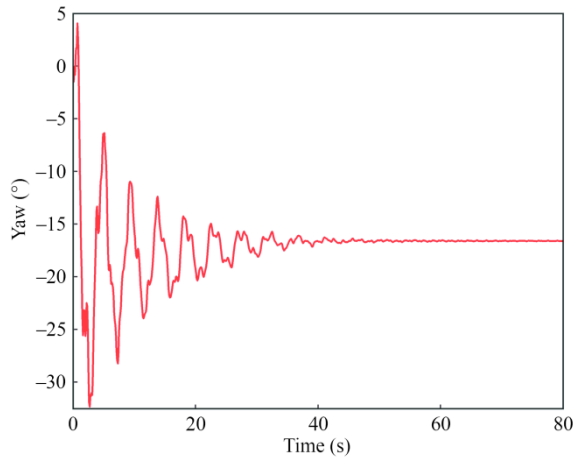


Fig. 26. Variation of yaw angle of the drogue( $V=170$  kt,  $H=2\ 286$  m).

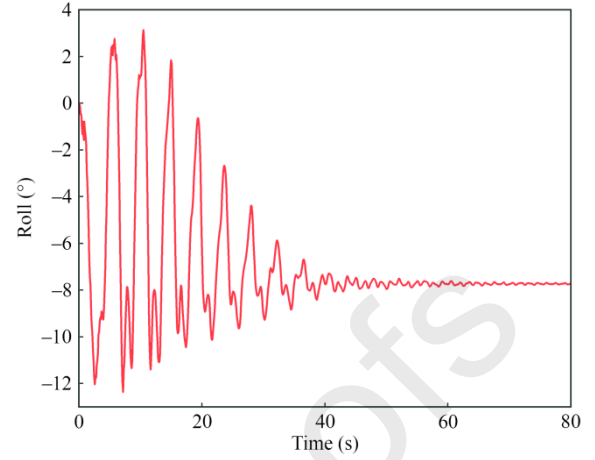


Fig. 27. Variation of roll angle of the drogue( $V=170$  kt,  $H=2\ 286$  m).

Fig. 28 shows the change of the displacement of underwing refueling and belly refueling in the  $Z$  direction (lateral) drogue relative to their initial points under the same working conditions ( $V=170$  kt,  $H=2\ 286$  m), and the red solid line is the displacement time history of the drogue of the underwing refueling mode, from which it can be seen that the displacement amplitude in the  $Z$  direction is relatively large, and the final stable position has a certain deviation from the starting point, as shown by the green dotted line, and is closer to the fuselage symmetry plane. The solid blue line is the displacement time history diagram of the belly refueling drogue, and its displacement in the  $Z$  direction is almost unchanged, which is due to the fact that the whole system is located in the symmetry plane of the fuselage, and the influence of the wake of the carrier aircraft cancels each other out on the symmetry plane, and has almost no effect on the system.

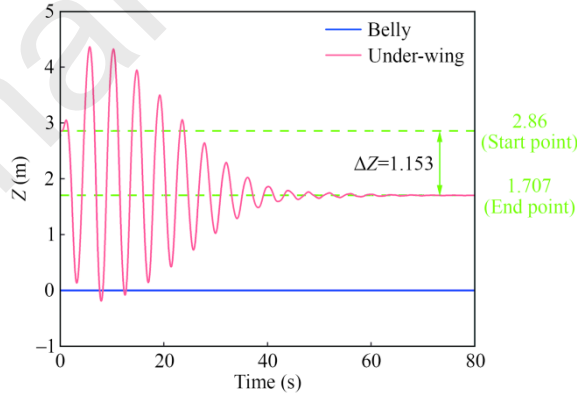


Fig. 28. Comparison of horizontal deflection displacement of underwing refueling

and belly refueling of drogue( $V=170$  kt,  $H=2\ 286$  m).

#### 4.2 Analysis of elongation release process of hose drogue system

The elongation release process of the hose-drogue system compared to the sagging process of the system in Section

4.1 involves the problem of variable mass and length of the boundary unit at the drag point, as mentioned before, the ALE description is introduced for this problem, and by combining with the ANCF method, the ALE-ANCF method is obtained, which will be used to perform the simulation analysis of this process. Therefore, the main content of this section is the simulation analysis of the strain release process of the hose-drogue system for different refueling positions. In this section, the time step for all the cases is 0.01 s, the total simulation time for each case is 50 s, and the system basically reaches equilibrium within 30 s, with an average CPU computation time of 70 s for each case. Fig. 29 shows the hose release velocity over time, with the total time of the elongation phase of 9.5 s, and the physical significance of the size of the area surrounded by the velocity curve and the time axis is the total length of the hose elongation.

In both refueling methods, the initial length of the hose is 1.64 m and both are placed horizontally as shown in Fig. 30 and Fig. 31. Figs. 32-34 illustrate the variation of each parameter of the system drogue with time for the abdominal refueling mode. Figs 35-40 illustrate the variation of each parameter of the system drogue with time for the under-wing refueling mode. It is worth noting that in the under-wing refueling mode, the drogue is in dynamic equilibrium and the center of mass of the drogue has small fluctuations of equal amplitude in the  $Z$  (lateral) direction from  $Z=2.65$  m, which may be due to the fact that the downwash and sidewash velocities at each position of the system are not the same as those at the other positions, and the fluctuations have a maximum value of about 10% of the total lateral displacement at equilibrium, so it is necessary to consider this factor in the design of the system. in order to mitigate the effects of the wake and thus improve the safety of the refueling system. In addition, by comparing the equilibrium position of the abdominal refueling method in Section 4.1 with the final stabilized position of the abdominal refueling method in this section, the stabilized positions of the two methods are highly consistent, which demonstrates the feasibility of the ALE-ANCF method for the simulation and analysis of the strain release process.

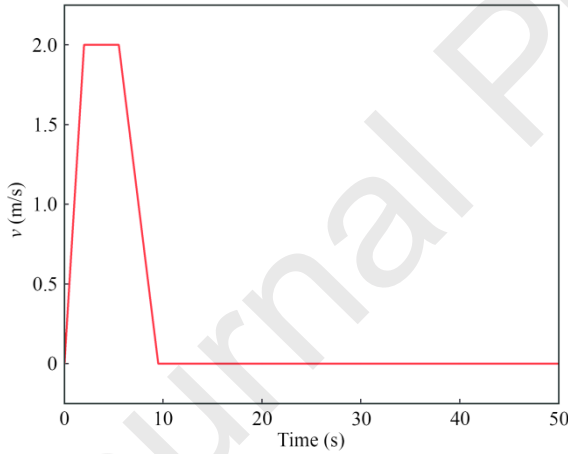


Fig. 29. Hose-drogue system release velocity vs time.

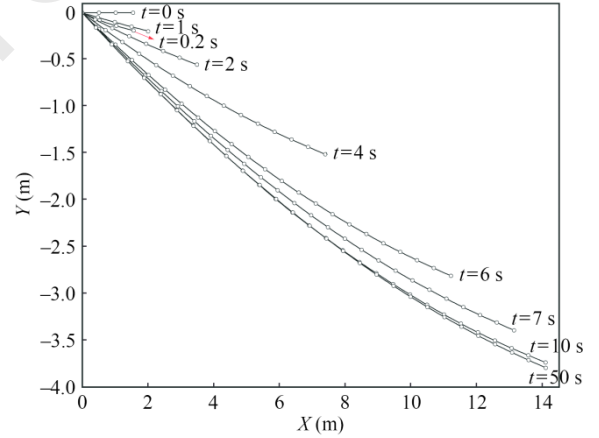


Fig. 30. Belly refueling system equilibrium process( $V=250$  kt,  $H=2\ 286$  m).

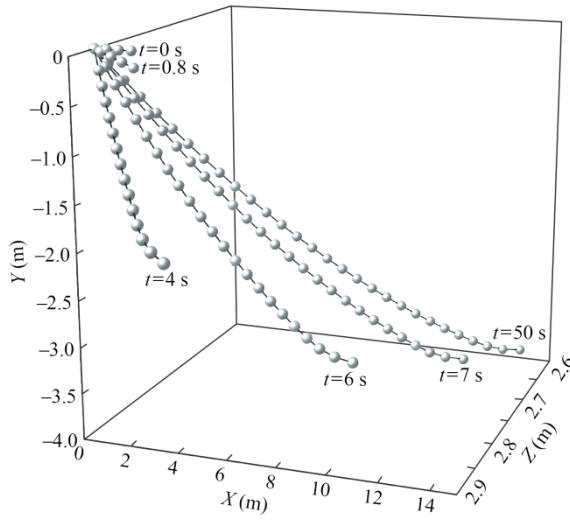


Fig. 31. Under-wing refueling system equilibrium process( $V=250$  kt,  $H=2\ 286$  m).

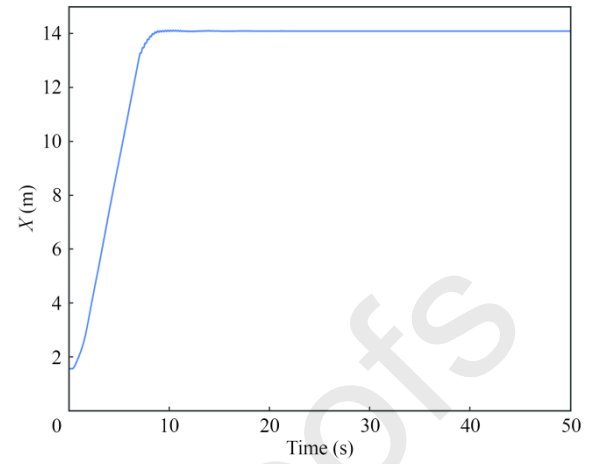


Fig. 32. Trajectory of the center of mass of drogue in  $X$ -direction for belly refueling( $V=250$  kt,  $H=2\ 286$  m).

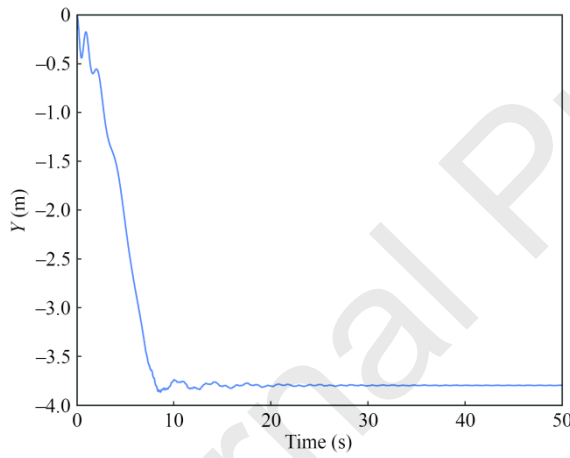


Fig. 33. Trajectory of the center of mass of drogue in  $Y$ -direction for belly refueling( $V=250$  kt,  $H=2\ 286$  m).

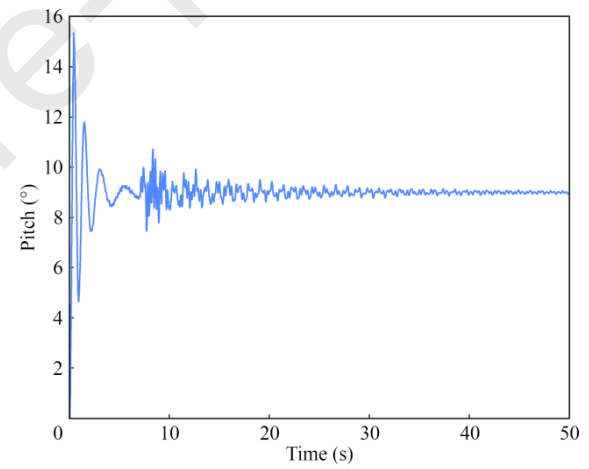


Fig. 34. Variation of pitch angle of drogue for belly refueling( $V=250$  kt,  $H=2\ 286$  m).



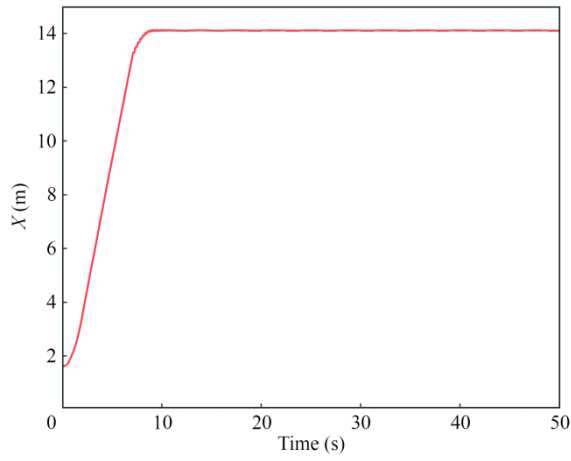


Fig. 35. Trajectory of the center of mass of drogue in  $X$ -direction for under-wing refueling ( $V=250$  kt,  $H=2\ 286$  m).

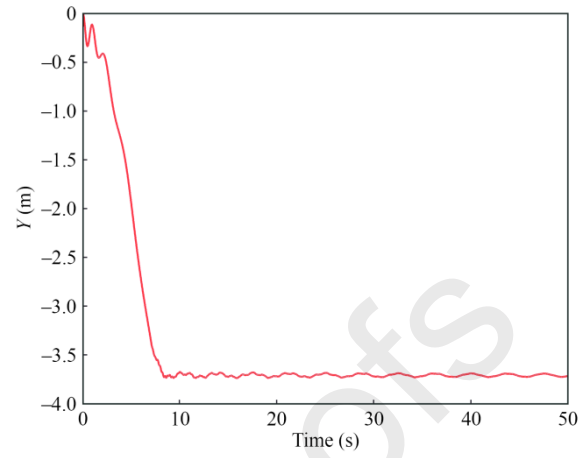


Fig. 36. Trajectory of the center of mass of drogue in  $Y$ -direction for under-wing refueling ( $V=250$  kt,  $H=2\ 286$  m).

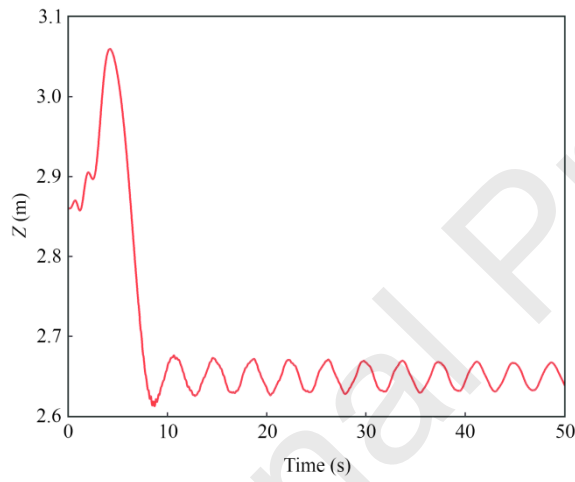


Fig. 37. Trajectory of the center of mass of drogue in  $Z$ -direction for under-wing refueling ( $V=250$  kt,  $H=2\ 286$  m).

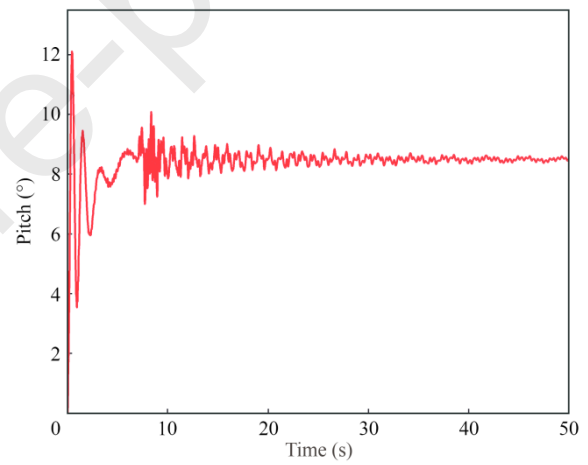


Fig. 38. Variation of the pitch angle of drogue for under-wing refueling ( $V=250$  kt,  $H=2\ 286$  m).

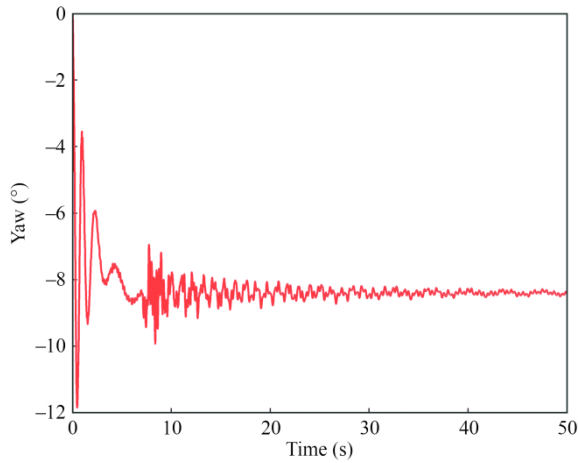


Fig. 39. Variation of yaw angle of drogue for under-wing refueling ( $V=250$  kt,  $H=2\ 286$  m).

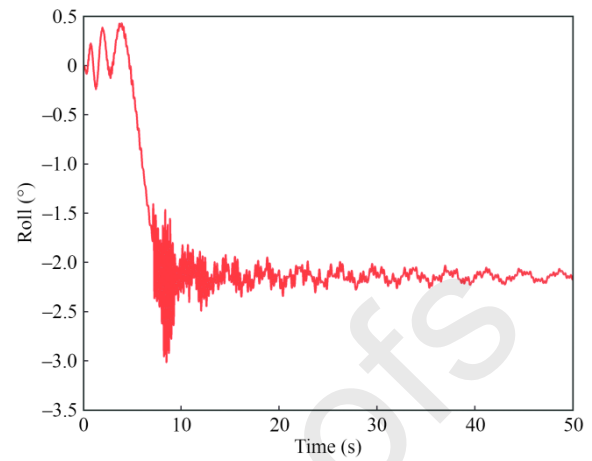


Fig. 40. Variation of roll angle of drogue for under-wing refueling ( $V=250$  kt,  $H=2\ 286$  m).

## 5. Conclusions

(1) The dynamic model of the aerial refueling hose-drogue system was established using the ANCF method. Under steady-state refueling conditions, the simulation results of the drogue position relative to the drag point are more consistent with the previously reported test flight results, while the accuracy under high-altitude conditions is further improved compared with the previous simulation results.

(2) Aiming at the elongation-release process of the air refueling hose-drogue system, based on the ALE description and combined with ANCF, a dynamics modeling method for solving the hose-drogue variable-length problem is finally proposed, and at the same time numerical simulations for different refueling scenarios are carried out to prove the universality of the method.

(3) The numerical solution program developed in this paper based on ANCF and ALE-ANCF methods for the rigid-flexible coupled multibody system of aerial refueling hose-drogue has made a breakthrough in computational efficiency while considering the accuracy, which can provide a qualitative reference for the design of the hose-drogue multibody system.

(4) The simulation results show that the dynamic characteristics of the lateral direction of the drogue are different due to the different effects of wake in different refueling methods, which is reflected in the whipping phenomenon of the hose-drogue in the actual aerial refueling process. This result can provide a reference for the design of underwing refueling systems to reduce the possibility of safety incidents.

(5) A discrepancy remains between the calculated and actual aerodynamic forces of the hose-drogue system, among other factors, which require further research. For instance, the impact of atmospheric turbulence has not been taken into account, and the empirical formulas utilized in the calculation of the aerodynamic forces are estimated. Additionally, the wake calculation employs the model summarized by previous authors.

## Acknowledgments

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## CRediT author statement

**Haizhen WANG:** Methodology, Software, Writing, Reviewing and Editing, Writing Original draft preparation. **Yang ZHANG:** Conceptualization, Data curation, Supervision. **Zhiwei SUN:** Visualization. **Bailing LIU:** Investigation. **Zhaobing JIANG:** Software, Validation.

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